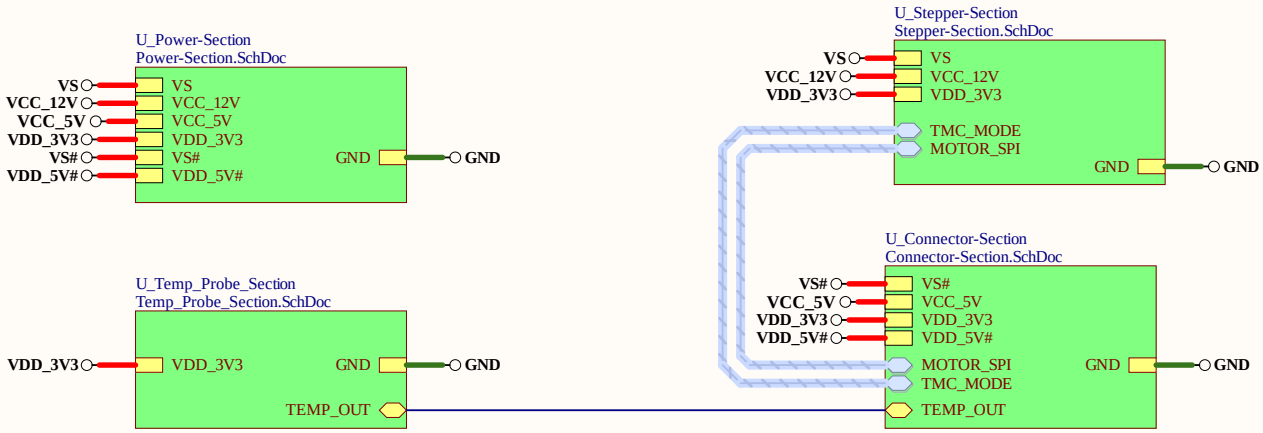
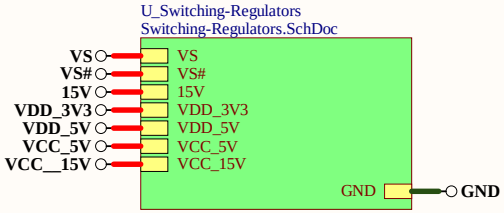
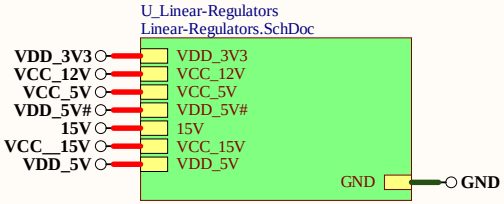
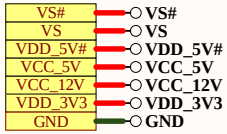
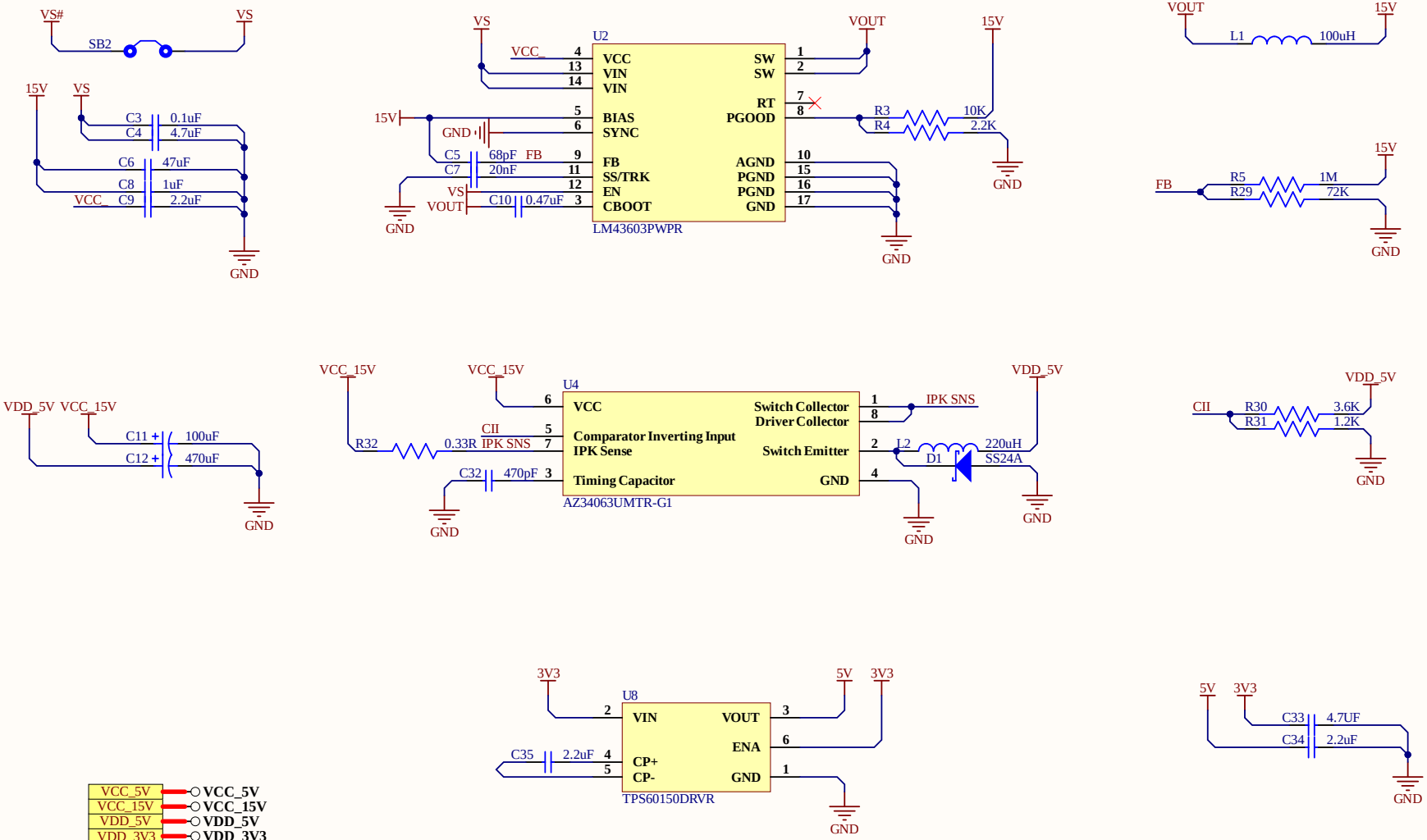
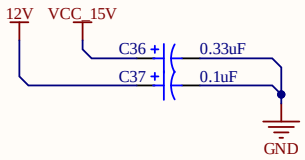

U_Power-Section
Power-Section.SchDoc
VS
VCC_12V
VCC_5V
VDD_3V3
VS#
VDD_5V#
VS
VCC_12V
VCC_5V
VDD_3V3
VS#
VDD_5V#
GND
GND
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VS
VCC_12V
VDD_3V3
TMC_MODE
MOTOR_SPI
GND
GND
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
VDD_3V3
GND
GND
TEMP_OUT
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VDD_5V#
VS#
VCC_5V
VDD_3V3
VDD_5V#
MOTOR_SPI
TMC_MODE
GND
GND
TEMP_OUT

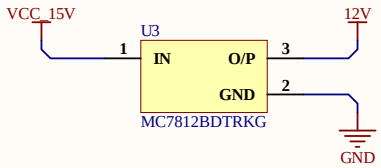

U_Switching-Regulators
Switching-Regulators.SchDoc
VS
VS#
15V
VDD_3V3
VDD_5V
VCC_5V
VCC_15V
GND
VS
VS#
15V
VDD_3V3
VDD_5V
VCC_5V
VCC_15V
GND
GND

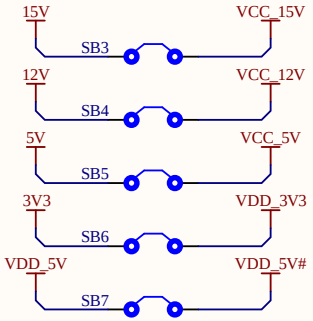

U_Linear-Regulators
Linear-Regulators.SchDoc
VDD_3V3
VCC_12V
VCC_5V
VDD_5V#
15V
VCC_15V
VDD_5V
GND
VDD_3V3
VCC_12V
VCC_5V
VDD_5V#
15V
VCC_15V
VDD_5V
GND
GND

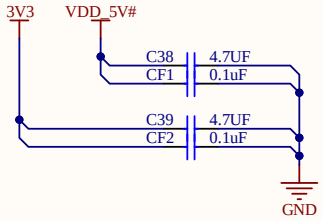

VS#
VS
VDD_5V#
VCC_5V
VCC_12V
VDD_3V3
GND
VS#
VS
VDD_5V#
VCC_5V
VCC_12V
VDD_3V3
GND

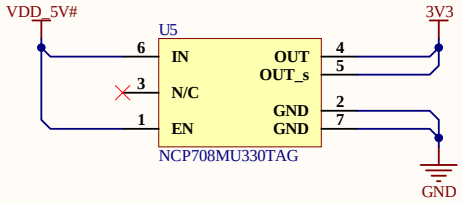

VS#
VS
SB2
15V
VS
C3 0.1uF
C4 4.7uF
C6 47uF
C8 1uF
VCC_ C9 2.2uF
GND
VS
VOUT
15V
U2
VCC_
VCC
VIN
VIN
BIAS
SYNC
FB
SS/TRK
EN
CBOOT
SW
SW
RT
PGOOD
AGND
PGND
PGND
GND
LM43603PWPR
C5 68pF FB
C7 20nF
VS
VOUT C10 0.47uF
R3 10K
R4 2.2K
GND
VOUT
15V
L1 100uH
FB
R5 1M
R29 72K
GND
VDD_5V
VCC_15V
C11 100uF
C12 470uF
GND
VCC_15V
VCC_15V
U4
VCC
Comparator Inverting Input
IPK Sense
Timing Capacitor
Switch Collector
Driver Collector
Switch Emitter
GND
AZ34063UMTR-G1
R32 0.33R IPK SNS
CII
C32 470pF
IPK SNS
L2 220uH
D1 SS24A
VDD_5V
GND
VDD_5V
CII
R30 3.6K
R31 1.2K
GND
3V3
5V
3V3
U8
VIN
CP+
CP-
VOUT
ENA
GND
TPS60150DRVR
C35 2.2uF
GND
5V
3V3
C33 4.7UF
C34 2.2uF
GND
VCC_5V
VCC_15V
VDD_5V
VDD_3V3
15V
VS#
VS
GND
VCC_5V
VCC_15V
VDD_5V
VDD_3V3
15V
VS#
VS
GND

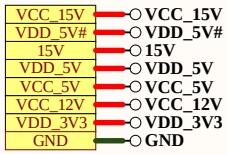

12V
VCC_15V
C36 +
0.33uF
C37 +
0.1uF
GND

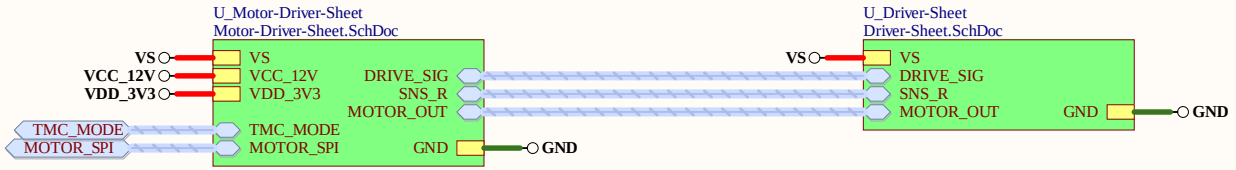

VCC_15V
U3
1
IN
O/P
3
GND
2
MC7812BDTRKG
12V
GND

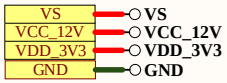

15V
VCC_15V
SB3
12V
VCC_12V
SB4
5V
VCC_5V
SB5
3V3
VDD_3V3
SB6
VDD_5V
VDD_5V#
SB7

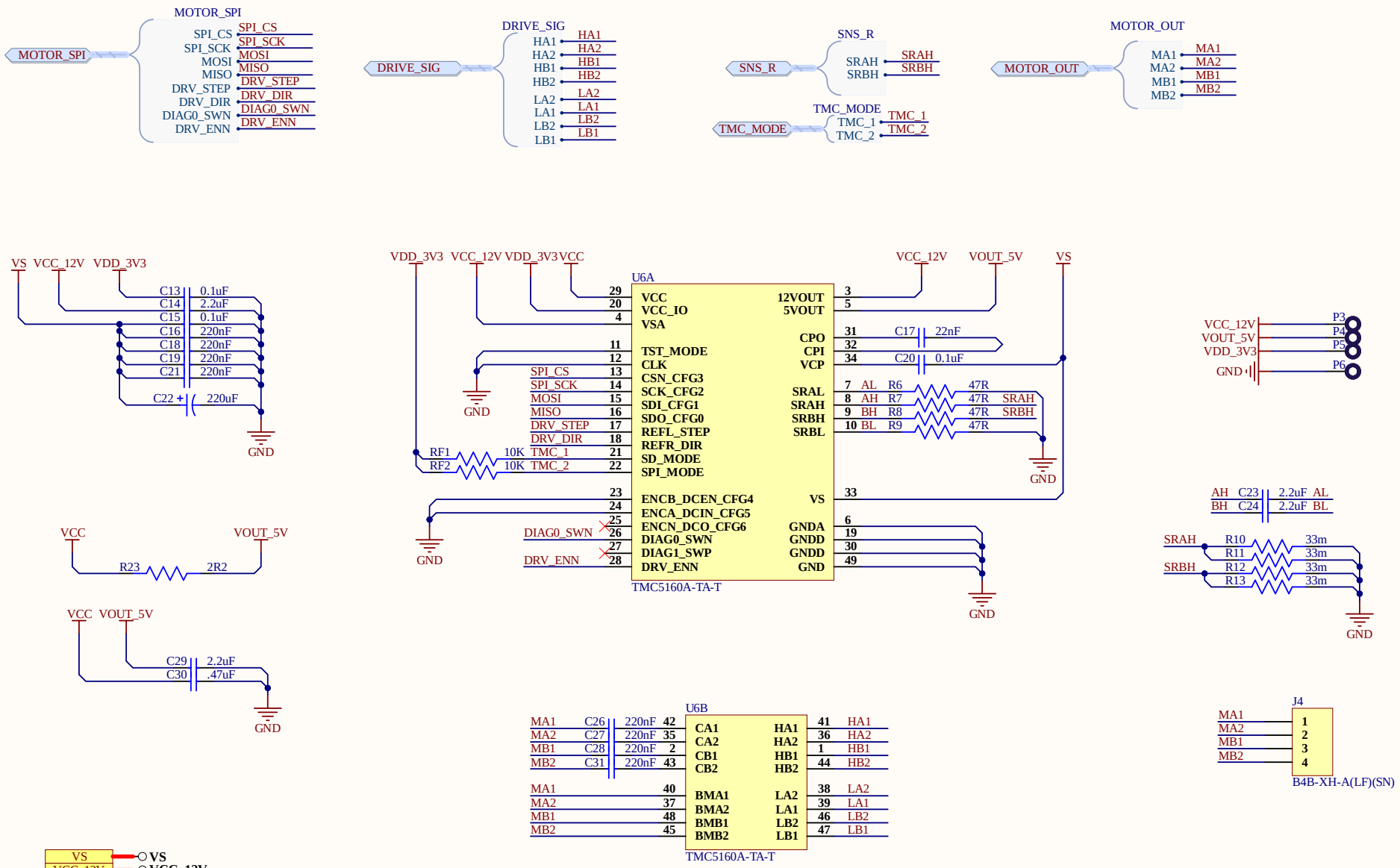

3V3
VDD_5V#
C38
4.7UF
CF1
0.1uF
C39
4.7UF
CF2
0.1uF
GND

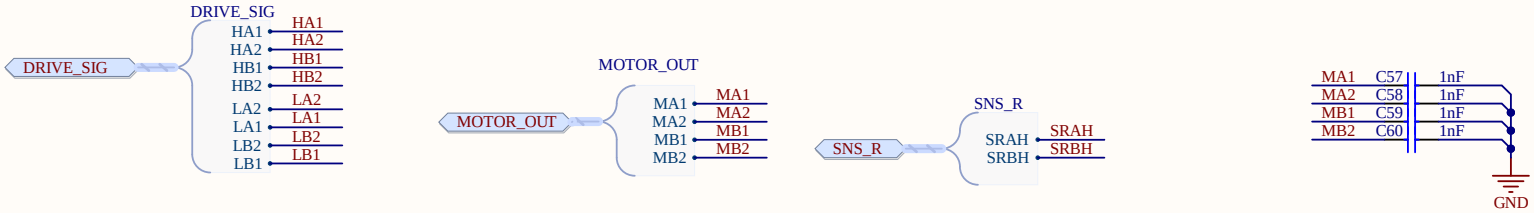

VDD_5V#
U5
6
IN
OUT
4
OUT_s
5
3
N/C
GND
2
1
EN
GND
7
NCP708MU330TAG
3V3
GND

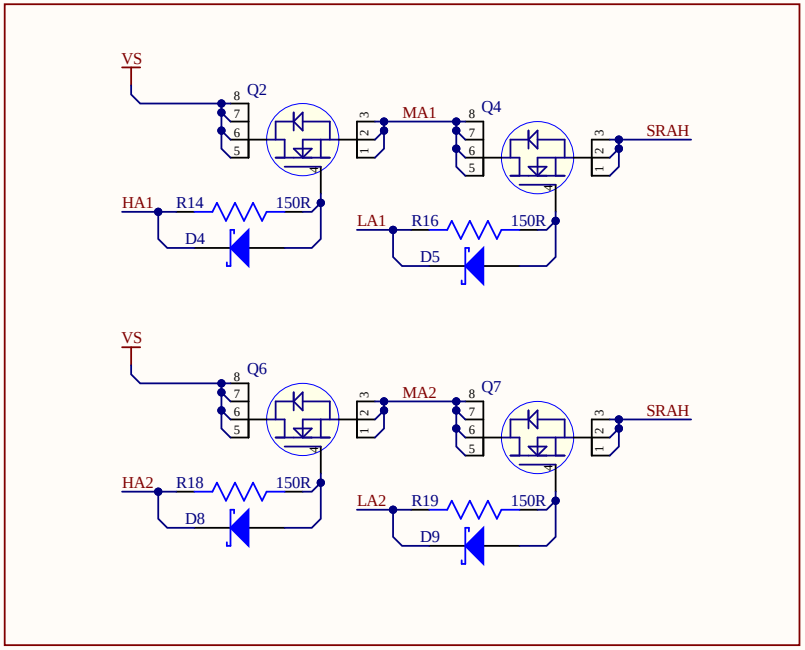

VCC_15V
VCC_15V
VDD_5V#
VDD_5V#
15V
15V
VDD_5V
VDD_5V
VCC_5V
VCC_5V
VCC_12V
VCC_12V
VDD_3V3
VDD_3V3
GND
GND

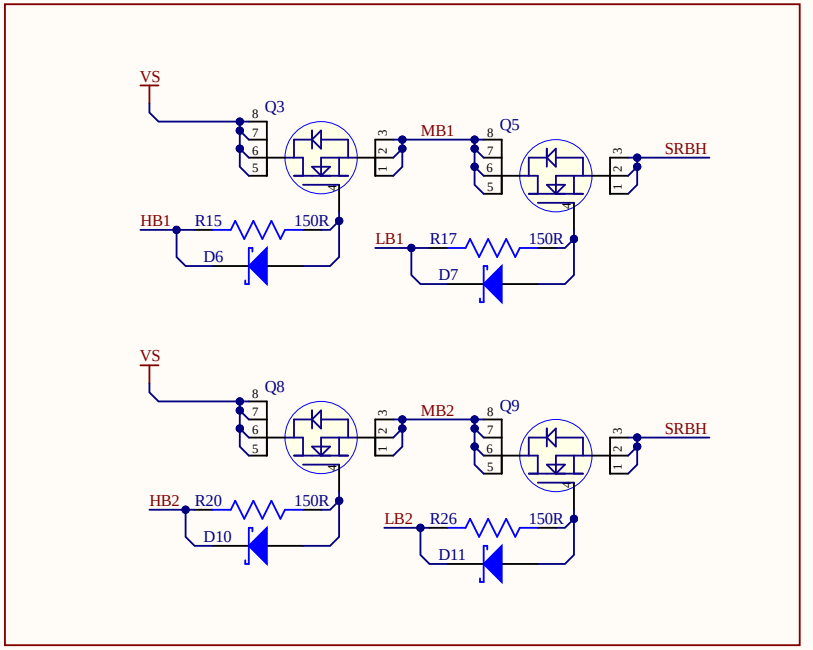

U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VDD_3V3
TMC_MODE
MOTOR_SPI
VS
VCC_12V
VDD_3V3
TMC_MODE
MOTOR_SPI
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND
U_Driver-Sheet
Driver-Sheet.SchDoc
VS
VS
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND


VS
VCC_12V
VDD_3V3
GND
VS
VCC_12V
VDD_3V3
GND


MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
SNS_R
SRAH
SRBH
TMC_MODE
TMC_1
TMC_2
MOTOR_OUT
MA1
MA2
MB1
MB2
VS
VCC_12V
VDD_3V3
GND
C13 0.1uF
C14 2.2uF
C15 0.1uF
C16 220nF
C18 220nF
C19 220nF
C21 220nF
C22 220uF
VCC
VOUT_5V
R23 2R2
C29 2.2uF
C30 .47uF
U6A
TMC5160A-TA-T
RF1 10K
RF2 10K
C17 22nF
C20 0.1uF
R6 47R
R7 47R
R8 47R
R9 47R
U6B
C26 220nF
C27 220nF
C28 220nF
C31 220nF
J4
B4B-XH-A(LF)(SN)
P3
P4
P5
P6
AH C23 2.2uF AL
BH C24 2.2uF BL
R10 33m
R11 33m
R12 33m
R13 33m

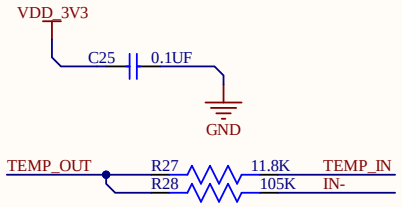

DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
MOTOR_OUT
MA1
MA2
MB1
MB2
SNS_R
SRAH
SRBH
C57 1nF
C58 1nF
C59 1nF
C60 1nF
GND

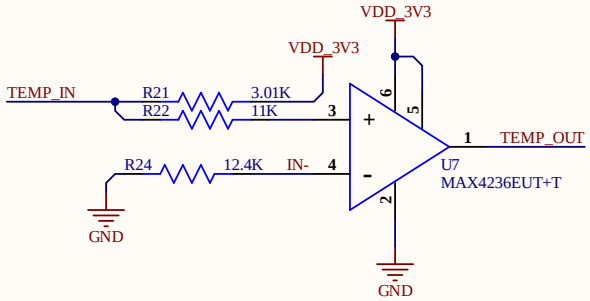

VS
Q2
MA1
Q4
SRAH
HA1
R14
150R
D4
LA1
R16
150R
D5
VS
Q6
MA2
Q7
SRAH
HA2
R18
150R
D8
LA2
R19
150R
D9

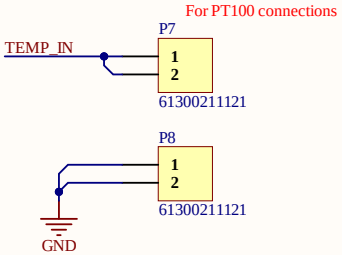

VS
Q3
MB1
Q5
SRBH
HB1
R15
150R
D6
LB1
R17
150R
D7
VS
Q8
MB2
Q9
SRBH
HB2
R20
150R
D10
LB2
R26
150R
D11

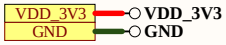

VS
GND
VS
GND

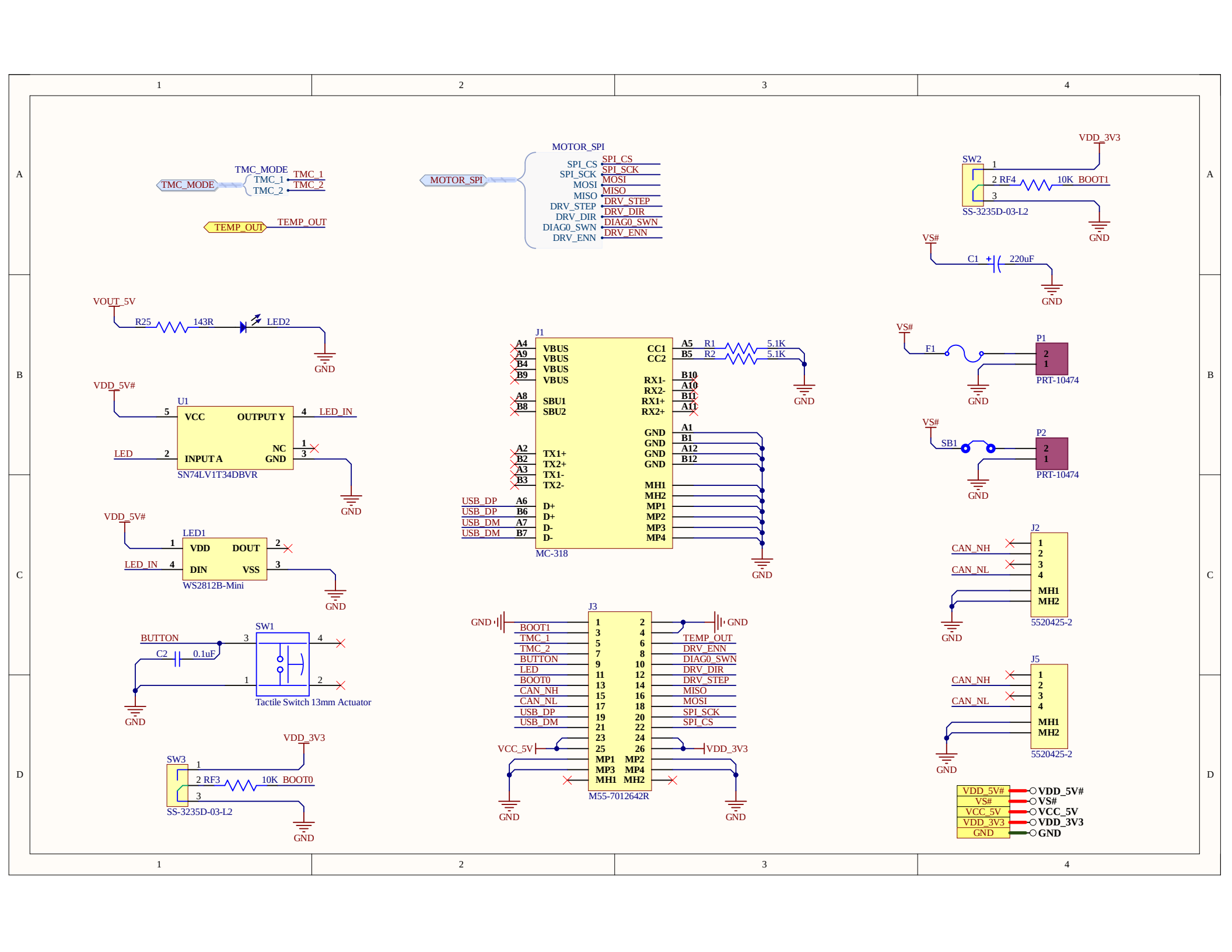

TEMP_OUT
TEMP_OUT


VDD_3V3
C25
0.1UF
GND
TEMP_OUT
R27
11.8K
TEMP_IN
R28
105K
IN-


VDD_3V3
VDD_3V3
TEMP_IN
R21
3.01K
R22
11K
3
6
5
+
1
TEMP_OUT
R24
12.4K
IN-
4
-
U7
MAX4236EUT+T
2
GND
GND


For PT100 connections
TEMP_IN
P7
1
2
61300211121
P8
1
2
61300211121
GND


VDD_3V3
GND
VDD_3V3
GND


TMC_MODE
TMC_1
TMC_2
TEMP_OUT
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
VDD_3V3
SW2
2 RF4
10K BOOT1
SS-3235D-03-L2
GND
VS#
C1
220uF
VOUT_5V
R25
143R
LED2
U1
VCC
OUTPUT Y
LED_IN
INPUT A
NC
SN74LV1T34DBVR
VDD_5V#
LED1
VDD
DOUT
DIN
VSS
WS2812B-Mini
J1
VBUS
SBU1
SBU2
TX1+
TX2+
TX1-
TX2-
D+
D-
CC1
CC2
RX1-
RX2-
RX1+
RX2+
GND
MH1
MH2
MP1
MP2
MP3
MP4
R1
R2
5.1K
USB_DP
USB_DM
MC-318
F1
P1
PRT-10474
SB1
P2
J2
CAN_NH
CAN_NL
5520425-2
J5
BUTTON
C2
0.1uF
SW1
Tactile Switch 13mm Actuator
J3
BOOT1
TMC_1
TMC_2
LED
BOOT0
CAN_NH
CAN_NL
USB_DP
USB_DM
TEMP_OUT
DRV_ENN
DIAG0_SWN
DRV_DIR
DRV_STEP
MISO
MOSI
SPI_SCK
SPI_CS
VCC_5V
M55-7012642R
SW3
2 RF3
10K BOOT0
VDD_5V#
VS#
VCC_5V
VDD_3V3
GND